Rmk : Another way to compute the solution is the method of *undetermined coefficients*.

The basic idea is to identify the formula for the parameters $\phi_1 \neq 0$ and $\phi_2 \neq 0$ in the postulated equation

$$C_t = \phi_1 A_t + \sum_{i=0}^{\infty} \phi_{2i} Y_{t+i}$$

When $Y_t = \bar{Y} \quad \forall t$, we can simplify into

$$C_t = \phi_1 A_t + \phi_2 Y_t$$

Assume that this consumption function satisfied both the Euler equation on consumption (optimal decisions) and the budget constraint (restrictions on the intertemporal allocation of consumption).
Can we find ϕ_1 ? and ϕ_2 ?